AI251P Applied Computer Vision

Capstone Project – Phase 1 Report

Camera-Trap Image Classification

Project 5 · Phase 1 (M1–M3, 40 points) · Due 28 October 2026

1 Team and project

Field	Your answer
Team name	Project 5
Topic	Project 5: camera-trap image classification
Application area	Wildlife monitoring and image triage
Code repository	https://github.com/salehdev3/Capstone_5
W&B project	https://forge.coreweave.com/wandb/saleh-alanazi-ministry-of-interior/camera-trap-phase1
Dataset card	https://github.com/salehdev3/Capstone_5/blob/v1/DATASET_CARD.md

Team members

Name	Student ID	Workstream (Table 3)	Phase 1 contribution
Saud Abdullah Alfaiz	225462	Data and classification	Data preparation, grouped split, audit and card
Saleh Mohammed Alanazi	225486	Data and classification	Scratch CNN and evaluation
Yousef Nafi Alanazi	225457	Data and classification	Frozen and full ResNet-50 comparison
Mohannad Mohammed Alqahtani	225461	Integration	Independent review, comparison and report

Problem and acceptance criteria

- **Summary:** The system classifies full camera-trap frames as empty or animal, then predicts animal species on human-box crops. It reports per-class metrics, calibration and errors.
- **Input → output:** A camera-trap photograph → empty/animal decision; an annotated animal crop → species category.
- **Intended users and the decision supported:** Wildlife-data reviewers use it to prioritize animal frames and inspect species predictions.
- **Acceptance criteria (fixed before results):** Animal recall $\geq$ .99; empty removal $\geq$ .20; species macro-F1 $\geq$ .65; species ECE $\leq$ .10.

2 Data foundations (M1) (12 points)

2.1 Datasets

Dataset	Link / ref.	Content and annotations	Use	Licence	Images
CCT20 subset	LILA CCT	Camera-trap frames and human boxes	All splits	CDLA-Permissive	6,022
Team-collected (bonus)	Private collection	Reviewed frames and boxes	Train/test	Team-owned; private	19

The notebook selected 6,022 CCT20 images under a fixed seed of 42, then included 19 reviewed team images. This project-specific subset does not use the official CCT benchmark split.

2.2 Cleaning

Check	Method / tool	Images removed
Corrupt or unreadable files	Download/decode checks	0
Exact duplicates	ID/hash checks	0
Near-duplicates	pHash cross-partition checks	0
Other (human audit quarantine)	One flagged event, four frames	4
Images remaining	6,041 initially kept; 6,037 after audit	6,037

2.3 Leakage-safe split

- **Grouping key:** CCT20 camera/event; local images by city and session.
- **Why this prevents leakage:** Frames from the same camera or burst share backgrounds, lighting and animals; grouping keeps these correlations within one partition.
- **Stratification and seed:** Group-aware approximate joint-label stratification, seed 42. Camera separation takes priority over exact class proportions.

Split	No. of Groups	No. of Images	Images per class (or % of the smallest class)
Train	13	3,534	Empty: 319; animal: 3,215
Validation selection	3	1,337	Empty: 91; animal: 1,246
Validation calibration	2	527	Empty: 13; animal: 514
Test	4	639	Empty: 53; animal: 586

Species crop support by split.

Species	Train	Selection	Calibration	Test
Badger	4 (0.1%)	0 (0.0%)	0 (0.0%)	1 (0.2%)
Bird	116 (4.1%)	3 (0.3%)	45 (9.3%)	9 (1.8%)
Bobcat	286 (10.1%)	25 (2.2%)	78 (16.1%)	57 (11.2%)
Cat	142 (5.0%)	247 (22.0%)	35 (7.2%)	48 (9.4%)
Coyote	288 (10.2%)	29 (2.6%)	37 (7.7%)	169 (33.3%)
Deer	11 (0.4%)	0 (0.0%)	0 (0.0%)	3 (0.6%)
Dog	117 (4.2%)	33 (2.9%)	73 (15.1%)	29 (5.7%)
Fox	1 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.2%)
Opossum	622 (22.1%)	489 (43.5%)	99 (20.5%)	87 (17.1%)
Rabbit	317 (11.2%)	164 (14.6%)	61 (12.6%)	4 (0.8%)
Raccoon	709 (25.2%)	14 (1.2%)	28 (5.8%)	28 (5.5%)
Rodent	38 (1.3%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Skunk	73 (2.6%)	0 (0.0%)	21 (4.3%)	19 (3.7%)
Squirrel	95 (3.4%)	120 (10.7%)	6 (1.2%)	53 (10.4%)
Total crops	2,819 (100%)	1,124 (100%)	483 (100%)	508 (100%)

Each cell reports crop count and its percentage within that split. A frame can contain multiple annotated animals; zero indicates no crop for that species.

2.4 Label audit

Audited sample size	Errors found	Estimated label-error rate	Upper bound (95% confidence)
100 unique training events	9 / 100 (9%)	9%	15.2% one-sided exact 95% upper bound

The audit file contains 120 reviewer records for 100 unique training images from 100 event IDs and 12 cameras. Twenty images received two reviews. The final adjudicated labels identify 9 incorrect labels (9%). The one-sided 95% Clopper–Pearson upper bound is 15.2%, calculated over 100 unique images rather than reviewer records. One image had different initial reviewer decisions and is represented once by its final adjudicated result. The binomial bound treats sampled events as independent; clustering within 12 cameras may make it optimistic. The audit is an annotation review; it does not by itself indicate that these labels were changed in the frozen training release.

2.5 Data loader and augmentation

RGB images were prepared with maximum side 1,024 pixels, resized to 224×224, then ImageNet-normalized. Training alone uses random flip and brightness/contrast augmentation from the shared preprocessing module; these variations reflect camera viewpoint and illumination. Exact augmentation parameters are fixed in `preprocessing.py`.

2.6 Dataset card summary

Raccoon and opossum dominate some groups while rare species have few examples; camera-group splits have distribution shift. CCT20 is largely US wildlife, and five held-out Tabuk frames cannot establish Saudi deployment performance. The audit covers 100 unique training events; its interval assumes independent sampled events. Source terms and course policy restrict image redistribution; no GPS or private photos are published. The full card is linked in Section 1.

3 Classification (M2–M3) (20 points)

Task: Full frames are classified empty/animal; human-box crops are classified by species. These classifiers provide the baseline for later detection and segmentation.

3.1 Runs compared

Six regimes were compared using validation-selection macro-F1; frozen ResNet-50 was chosen for both tasks. The locked test set was evaluated after selection. W&B links for all six runs are in the linked project.

Run (W&B link)	Backbone	Training	Input res.	Epochs	Val. acc.	Val. macro-F1
Triage (W&B)	CNN	From scratch	224	6	.941	.601
Triage (W&B)	ResNet-50	Pretrained, frozen	224	12	.940	.662
Triage (W&B)	ResNet-50	Pretrained, full fine-tuning	224	12	.927	.640
Species (W&B)	CNN	From scratch	224	12	.505	.156
Species (W&B)	ResNet-50	Pretrained, frozen	224	12	.761	.364
Species (W&B)	ResNet-50	Pretrained, full fine-tuning	224	12	.750	.340

Test-set results (final models only)

Run	Test accuracy (95% CI)	Test macro-F1	ECE
Triage scratch	.917 [.868,.966]	.478	.0463
Triage frozen (selected)	.933 [.897,.967]	.652	.0460
Species scratch	.350 [.284,.386]	.178	.0733
Species frozen (selected)	.742 [.642,.876]	.482	.0568

Which model you keep and why: Frozen ResNet-50 gave the highest validation macro-F1 for both tasks (.662 triage and .364 species). It was therefore retained before test results were examined. At test threshold .3416, triage retained 586/586 animal frames but removed only 6/53 empty frames (11.3%), missing the 20% empty-removal target. Species test macro-F1 was .482, below the .65 target; ECE .0568 met the .10 limit.

3.2 Learning curves

Species full fine-tuning reached 94.7% training accuracy but 75.0% validation accuracy, consistent with overfitting. Frozen species reached 89.7% and 76.1%, respectively. Rare-class support and camera shift remain important limits.

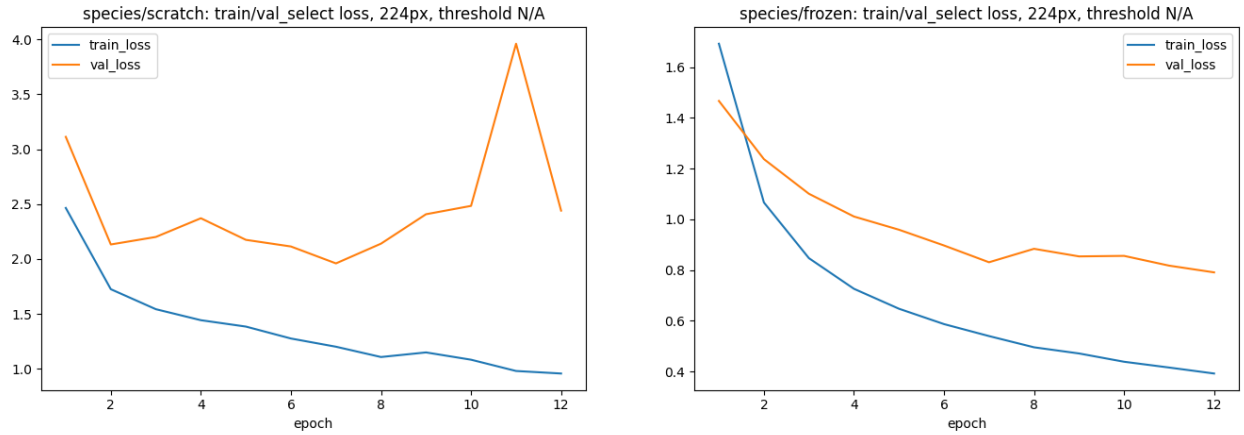


Figure 1: Training and validation-selection loss: (left) scratch CNN, (right) frozen ResNet-50. Split: train and validation-selection; input: 224×224; threshold: argmax.

3.3 Per-class results (selected model, test set)

Class	Precision	Recall	F1	Support
Empty (triage)	.917	.208	.338	53
Animal (triage)	.933	.998	.965	586
Badger (species)	.000	.000	.000	1
Bird (species)	.900	1.000	.947	9
Bobcat (species)	.629	.684	.655	57
Cat (species)	.632	.250	.358	48
Coyote (species)	.956	.763	.849	169
Deer (species)	.000	.000	.000	3
Dog (species)	.676	.862	.758	29
Fox (species)	.000	.000	.000	1
Opossum (species)	.656	.966	.781	87
Rabbit (species)	.250	.250	.250	4
Raccoon (species)	.366	.536	.435	28
Rodent (species)	.000	.000	.000	0
Skunk (species)	.933	.737	.824	19
Squirrel (species)	.860	.925	.891	53
Macro average (species)	.490	.498	.482	508

Top-k accuracy (species, $k = 3$): .9055 on test; 224×224, argmax.

3.4 Confusion matrix and calibration

Triage confusion matrix.

Test set ($n = 639$; input 224×224). The threshold 0.3416 was selected on the calibration split (`val_calibrate`, $n = 527$) to target animal recall ≥ 0.99, after temperature scaling ($T = 0.714$). The test set was not used to select the threshold.

	Argmax		Threshold 0.3416	
True class	Empty	Animal	Empty	Animal
Empty ($n = 53$)	11	42	6	47
Animal ($n = 586$)	1	585	0	586

The per-class results above use argmax (empty recall .208). At the selected operating threshold, test animal recall was 1.000 and empty removal was $6/53 = 11.3\%$.

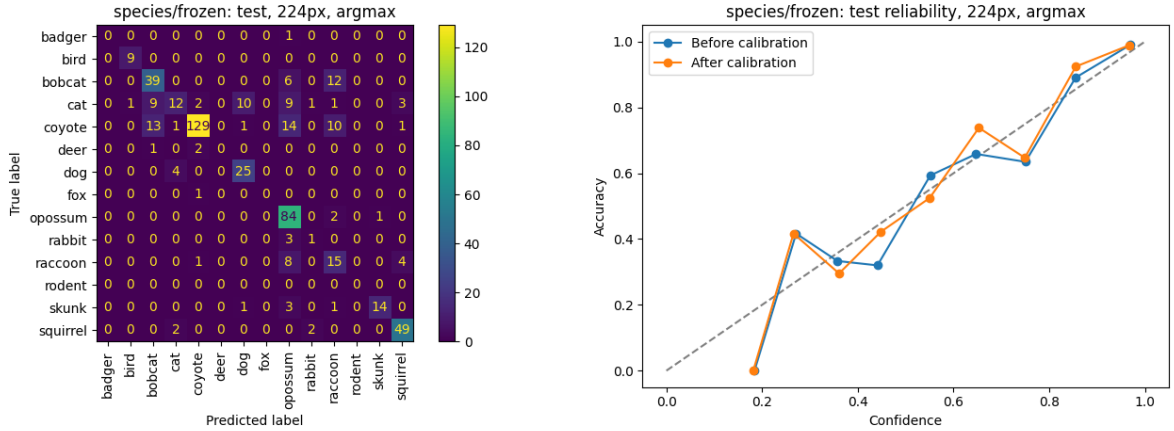


Figure 2: (Left) confusion matrix; (right) reliability diagram. Split: test; input: 224×224; threshold: argmax.

Bobcat was repeatedly confused with raccoon in the gallery; a fox crop was called coyote. Selected species test ECE was .0568 versus raw .0454, so temperature scaling did not improve ECE on this test split. Gallery observations are not a causal proof.

3.5 Error analysis

Full-test error counts. The selected frozen ResNet-50 produced 43 triage argmax errors among 639 frames and 131 species errors among 508 crops. Species errors comprise 32 distinct true-to-predicted class pairs. The table lists the most frequent pairs; the complete counts are archived in `evidence/full_test_error_patterns.csv`. The gallery shows selected examples only.

Task	True class → predicted class	Full-test errors
Triage (argmax)	Empty → animal	42
Triage (argmax)	Animal → empty	1
Species (argmax)	Coyote → opossum	14
Species (argmax)	Coyote → bobcat	13
Species (argmax)	Bobcat → raccoon	12
Species (argmax)	Cat → dog	10
Species (argmax)	Coyote → raccoon	10

Grayscale proxy analysis. The manifest's grayscale flag is a visual proxy and is not a verified day/night label. Error rates below use argmax predictions and include all samples with a matching manifest record.

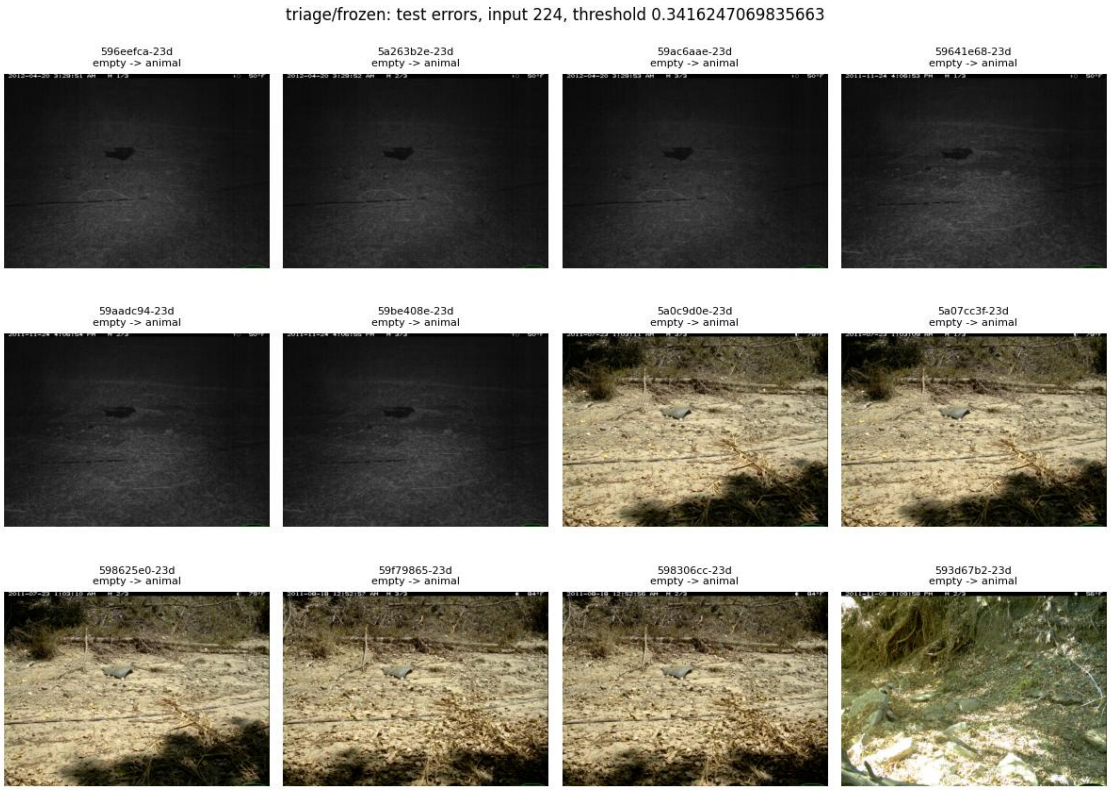


Figure 3: Triage test error gallery: recorded label and prediction are shown; input 224×224, animal threshold .3416. Several source labels need human verification.

Task	Grayscale proxy	Samples	Errors	Error rate
Triage	No	304	35	11.5%
Triage	Yes	335	8	2.4%
Species crops	No	246	42	17.1%
Species crops	Yes	262	89	34.0%

Triage frame predictions were matched to the image manifest; species crop predictions inherited the proxy flag from their source frame. These associations do not establish nighttime error rates or a causal lighting effect. Complete error counts and source-image proxy summaries accompany this report.

3.6 Data-improvement plan

Finding from the error analysis	Planned action	Expected effect
Possible source-label errors	Review full-resolution originals; report corrected-label sensitivity separately	Distinguish annotation issues from model errors
Rare species and empty frames	Add verified, independently grouped examples; tune on validation only	Improve minority-class recall and empty removal
Potential illumination effect	Record human-verified day/night labels for training and validation images; compare illumination-balanced training on validation groups	Measure nighttime recall without treating grayscale appearance as ground truth

4 Reproducibility (8 points)

Item	Your answer
Repository and commit hash	https://github.com/salehdev3/Capstone_5, evaluated results release `2cc1731` on `v1`
Configuration file used for the reported runs	`handoff/data_config.json` and `runs/member2,member3/config.json`
Random seeds	42
Requirements file (pinned versions)	`requirements-colab-2026-10-28.txt`; Python 3.13.16, Linux x86_64; PyTorch 2.11.0+cu130; torchvision 0.26.0+cu130
Command(s) to reproduce the main result	Run `notebooks/Phase1_Final_Presentation.ipynb` in section order on Colab GPU
Where the data is obtained (not committed)	CCT20 on LILA; private team photos; recorded manifest hash in dataset card
Reproduced by (team member) and result	Mohannad, 28 October 2026; assessor feedback reports matching run metrics. Fresh-process inference results are archived separately

The assessor reported that Mohannad's full rerun on 28 October 2026 matched the recorded metrics. The following side-by-side values reproduce that reported agreement for the six model/task combinations. W&B links for the recorded runs are listed in the experiment results.

Run	Metric	Original	28 Oct rerun
Triage / scratch	Accuracy	.917	.917
	Macro-F1	.478	.478
	ECE	.046	.046
Species / scratch	Accuracy	.350	.350
	Macro-F1	.178	.178
	ECE	.073	.073
Triage / frozen	Accuracy	.933	.933
	Macro-F1	.652	.652
	ECE	.046	.046
Triage / full	Accuracy	.909	.909
	Macro-F1	.594	.594
	ECE	.049	.049
Species / frozen	Accuracy	.742	.742
	Macro-F1	.482	.482
	ECE	.057	.057
Species / full	Accuracy	.764	.764
	Macro-F1	.455	.455
	ECE	.061	.061

5 Submission checklist

☐ Corrupt files and duplicates handled, with counts reported

☐ Split is grouped by the source of correlation; the grouping key is justified

☐ Label audit reported with a confidence bound

☐ Dataset card states licences, limitations and privacy measures

☐ Scratch and fine-tuned classifiers compared in W&B

☐ Per-class metrics, confusion matrix and calibration (ECE) reported

☐ Test set used only for final evaluation, not model selection

☐ Data-improvement plan follows from the error analysis

☐ Fixed seeds, recorded configuration and pinned Colab environment; no hard-coded paths

☐ Every figure states split, input resolution and thresholds

☐ No data or personal information committed to the repository

6 Self-assessment

Criterion	Points	Self-score	Strongest evidence (section)	Known gap
Data foundations and dataset card (M1)	12	11	Cleaning, split, audit, card	Small initial audit
Classification (M2–M3)	20	19	Six runs, metrics and galleries	Two preset targets missed
Reproducibility	8	8	Pinned Colab environment, checkpoint hashes and Mohannad replay comparison	
Total	**40**	**38**	Team self-assessment	Instructor decides final score

7 Obstacles and how they were addressed

A Colab upload stall interrupted evaluation. The evaluation workflow now saves test metrics before a time-bounded W&B upload, so a stalled upload does not block the local results.

Obstacle	Module	Impact on the work	Action taken	Status
Colab upload stalls	M2–M3	Evaluation interruption	Save metrics before bounded W&B upload	Resolved